

STA623 - Bayesian Data Analysis

Practical 5: the Bayesian workflow: regression, model checking, comparison and hierarchical models (with solutions)

Marc Henrion

2026-10-02

Goals. Carry out a complete Bayesian analysis: prior predictive checks, fitting, diagnostics, interpretable summaries, posterior predictive checks, model comparison with LOO, and hierarchical modelling with partial pooling. Accompanies Lecture 5.

The helper `loo_from_ll()` from the lecture notes is defined in the (hidden) setup chunk.

Exercise 1: logistic regression, from priors to predictions

A cross-sectional study of 400 adults attending an outpatient clinic records hypertension (yes/no), age and sex:

```
set.seed(51)
N <- 400
bp <- tibble(age = round(runif(N, 18, 75)), male = rbinom(N, 1, 0.45)) |>
  mutate(hyp = rbinom(N, 1, plogis(-1.2 + 0.6 * (age - 45) / 10 + 0.3 * male)),
         age10_c = (age - 45) / 10)
```

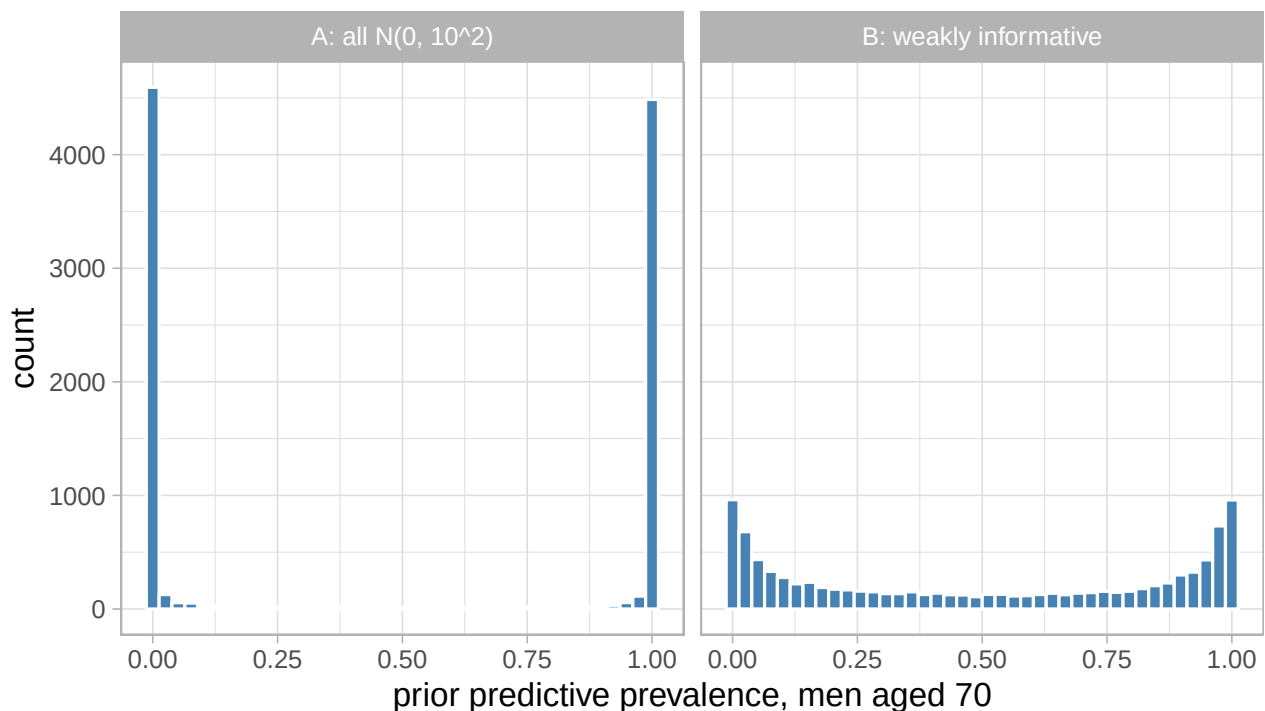
Model: $\text{logit } P(\text{hyp}_i = 1) = \beta_0 + \beta_1 \text{age10_c}_i + \beta_2 \text{male}_i + \beta_3 \text{age10_c}_i \times \text{male}_i$.

- Prior predictive check.** Compare two prior sets: (A) all coefficients $\sim \mathcal{N}(0, 10^2)$; (B) $\beta_0 \sim \mathcal{N}(0, 1.5^2)$, $\beta_1, \beta_2 \sim \mathcal{N}(0, 1)$, $\beta_3 \sim \mathcal{N}(0, 0.5^2)$. For each, simulate the prior predictive distribution of the *prevalence of hypertension among 70-year-old men*. Which prior set is more sensible?
- Fit the model with prior set B in NIMBLE (4 chains). Check convergence.
- Report posterior medians and 95% credible intervals for the odds ratio per 10 years of age in women, the odds ratio per 10 years in men, and the male-vs-female odds ratio at age 45. What is the posterior probability that the effect of age is stronger in men than in women?
- Plot predicted prevalence by age for men and women with 95% credible bands.

Solution

(a)

```
S <- 1e4
x70 <- (70 - 45) / 10
prev70 <- function(sd0, sd1, sd2, sd3) {
  b <- cbind(rnorm(S, 0, sd0), rnorm(S, 0, sd1), rnorm(S, 0, sd2), rnorm(S, 0,
    ↪ sd3))
  plogis(b[, 1] + b[, 2] * x70 + b[, 3] + b[, 4] * x70)
}
tibble(`A: all N(0, 10^2)` = prev70(10, 10, 10, 10),
      `B: weakly informative` = prev70(1.5, 1, 1, 0.5)) |>
  pivot_longer(everything()) |>
  ggplot(aes(value)) + geom_histogram(bins = 40, fill = "steelblue", colour =
    ↪ "white") +
  facet_wrap(~name) + labs(x = "prior predictive prevalence, men aged 70")
```



Prior set A says the prevalence among 70-year-old men is almost certainly either about 0% or about 100%, which nobody believes. Prior set B spreads its mass across the whole range while favouring neither extreme.

(b)

```
hyp_code <- nimbleCode({
  for (i in 1:N) {
    hyp[i] ~ dbern(p[i])
    logit(p[i]) <- b0 + b1 * age[i] + b2 * male[i] + b3 * age[i] * male[i]
  }
  b0 ~ dnorm(0, sd = 1.5)
  b1 ~ dnorm(0, sd = 1)
  b2 ~ dnorm(0, sd = 1)
  b3 ~ dnorm(0, sd = 0.5)
})
```

```
hyp_fit <- nimbleMCMC(hyp_code,
  constants = list(N = N, age = bp$age10_c, male = bp$male), data = list(hyp =
    ↪ bp$hyp),
  inits = function() list(b0 = rnorm(1), b1 = rnorm(1), b2 = rnorm(1), b3 =
    ↪ rnorm(1, 0, 0.3)),
  monitors = c("b0", "b1", "b2", "b3"), nchains = 4, niter = 6000, nburnin = 1000,
  setSeed = TRUE, progressBar = FALSE, samplesAsCodaMCMC = TRUE)
```

```
summarise_draws(as_draws(hyp_fit), "median", ~quantile(.x, c(0.025, 0.975)),
  ↪ "rhat", "ess_bulk", "ess_tail")
```

```
# A tibble: 4 x 7
  variable median `2.5%` `97.5%` rhat ess_bulk ess_tail
  <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 b0        -0.985 -1.33  -0.660 1.00   1327. 2199.
2 b1         0.558  0.382  0.771 1.00   1378. 2099.
3 b2         0.304 -0.158  0.778 1.00   1191. 2349.
4 b3        -0.0204 -0.311  0.252 1.00   1349. 2438.
```

All $\widehat{R} < 1.01$ and ESS well above 400.

(c)

```
d <- as.matrix(hyp_fit)
qs <- function(x) quantile(x, c(0.5, 0.025, 0.975))
rbind(`OR per 10y, women` = qs(exp(d[, "b1"])),
  `OR per 10y, men` = qs(exp(d[, "b1"] + d[, "b3"])),
  `OR male vs female at 45` = qs(exp(d[, "b2"]))) |> kable(digits = 3)
```

	50%	2.5%	97.5%
OR per 10y, women	1.747	1.465	2.163
OR per 10y, men	1.716	1.395	2.121

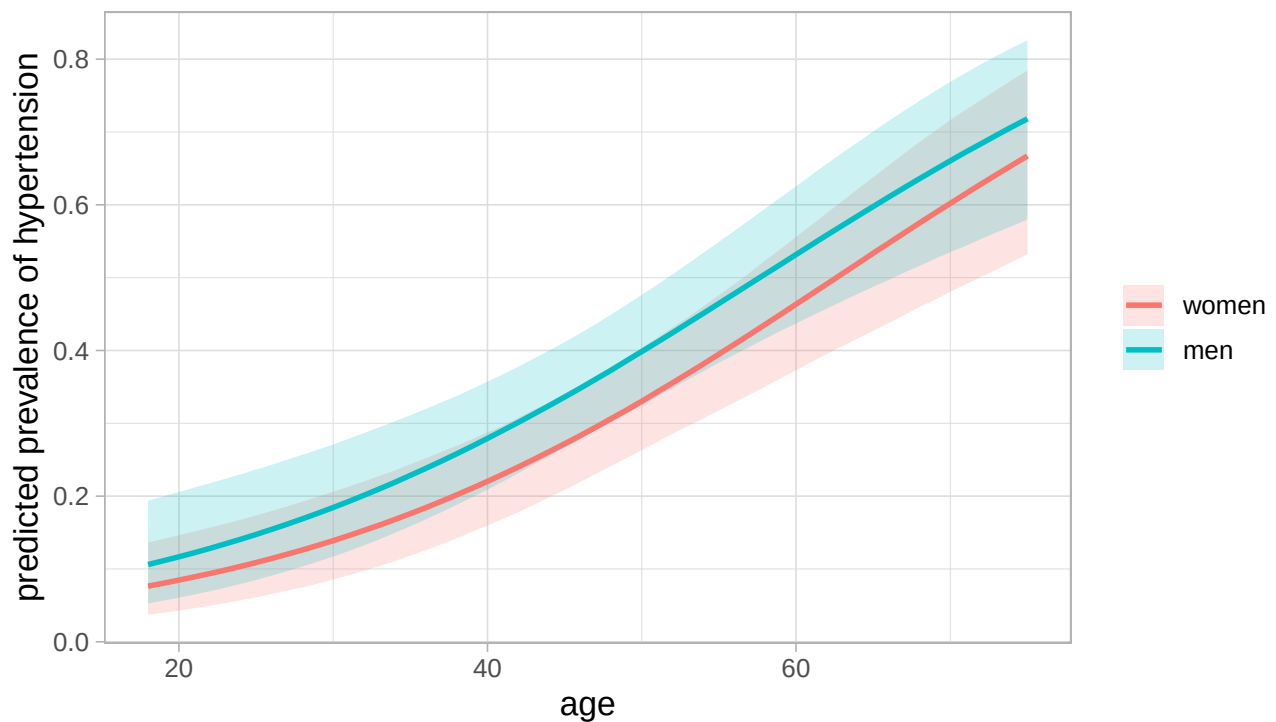
	50%	2.5%	97.5%
OR male vs female at 45	1.356	0.854	2.178

```
c(`P(age effect stronger in men)` = mean(d[, "b3"] > 0))
```

```
P(age effect stronger in men)
0.44115
```

(d)

```
pred <- expand_grid(age = 18:75, male = 0:1) |>
  mutate(draws = map2(age, male, ~ plogis(d[, "b0"] + d[, "b1"] * (.x - 45) / 10
    ↪ + d[, "b2"] * .y +
                                     d[, "b3"] * (.x - 45) / 10 * .y)),
    med = map_dbl(draws, median),
    lo = map_dbl(draws, ~ quantile(.x, 0.025)), hi = map_dbl(draws, ~
    ↪ quantile(.x, 0.975)),
    sex = factor(male, labels = c("women", "men")))
ggplot(pred, aes(age, med, colour = sex, fill = sex)) +
  geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2, colour = NA) +
  ↪ geom_line(linewidth = 1) +
  labs(y = "predicted prevalence of hypertension", colour = NULL, fill = NULL)
```



Exercise 2: posterior predictive checks and LOO for count data

Number of outpatient visits in one year for 250 people living with diabetes, by whether they live more than 10 km from the clinic:

```
set.seed(52)
M <- 250
visits_df <- tibble(far = rbinom(M, 1, 0.4)) |>
  mutate(visits = rnbinom(M, size = 2, mu = exp(1.8 - 0.4 * far)))
visits_df |> group_by(far) |> summarise(n = n(), mean = mean(visits), var =
  ↪ var(visits))
```

```
# A tibble: 2 x 4
  far     n mean  var
<int> <int> <dbl> <dbl>
1     0  143  5.53  22.2
2     1  107  3.93  11.0
```

- (a) Fit a Poisson regression and a negative binomial regression ($\log \mu_i = \alpha_0 + \alpha_1 \text{far}_i$; use $\alpha_0 \sim \mathcal{N}(1, 1.5^2)$, $\alpha_1 \sim \mathcal{N}(0, 1)$, and for the negative binomial a dispersion $\phi \sim \text{Exp}(1)$ with size $r = 1/\phi$).
- (b) Compare the posterior of the rate ratio e^{α_1} under the two models.
- (c) Posterior predictive check: for each model simulate 1,000 replicated datasets (each from a different posterior draw). Use as test statistics the standard deviation of visits, the proportion of people with 0 or 1 visits, and the difference in mean visits between groups. Compute posterior predictive p-values.
- (d) Compare the models with LOO. Which would you report and why?

Solution

(a)

```
pois_code <- nimbleCode({
  for (i in 1:N) { y[i] ~ dpois(mu[i]); log(mu[i]) <- a0 + a1 * far[i] }
  a0 ~ dnorm(1, sd = 1.5); a1 ~ dnorm(0, sd = 1)
})
nb_code <- nimbleCode({
  for (i in 1:N) {
    y[i] ~ dnegbin(prob[i], r)
    prob[i] <- r / (r + mu[i])
    log(mu[i]) <- a0 + a1 * far[i]
  }
  a0 ~ dnorm(1, sd = 1.5); a1 ~ dnorm(0, sd = 1)
  phi ~ dexp(1); r <- 1 / phi
})
```

```

cst <- list(N = M, far = visits_df$far); dat <- list(y = visits_df$visits)
fit_p <- nimbleMCMC(pois_code, constants = cst, data = dat,
  inits = function() list(a0 = rnorm(1, 1, 0.5), a1 = rnorm(1, 0, 0.5)),
  monitors = c("a0", "a1"), nchains = 4, niter = 5000, nburnin = 1000,
  setSeed = TRUE, progressBar = FALSE, samplesAsCodaMCMC = TRUE)
fit_nb <- nimbleMCMC(nb_code, constants = cst, data = dat,
  inits = function() list(a0 = rnorm(1, 1, 0.5), a1 = rnorm(1, 0, 0.5), phi =
    ↪ runif(1, 0.2, 1)),
  monitors = c("a0", "a1", "phi"), nchains = 4, niter = 5000, nburnin = 1000,
  setSeed = TRUE, progressBar = FALSE, samplesAsCodaMCMC = TRUE)

```

```

summarise_draws(as_draws(fit_p), "median", ~quantile(.x, c(0.025, 0.975)),
  ↪ "rhat", "ess_bulk")

```

```

# A tibble: 2 x 6
  variable median `2.5%` `97.5%` rhat ess_bulk
  <chr>      <dbl> <dbl> <dbl> <dbl> <dbl>
1 a0         1.71  1.64  1.78  1.00  1699.
2 a1        -0.341 -0.456 -0.228 1.00  1776.

```

```

summarise_draws(as_draws(fit_nb), "median", ~quantile(.x, c(0.025, 0.975)),
  ↪ "rhat", "ess_bulk")

```

```

# A tibble: 3 x 6
  variable median `2.5%` `97.5%` rhat ess_bulk
  <chr>      <dbl> <dbl> <dbl> <dbl> <dbl>
1 a0         1.71  1.58  1.85  1.00  1764.
2 a1        -0.344 -0.551 -0.115 1.00  1742.
3 phi         0.532  0.408  0.691 1.00  3612.

```

(b)

```

dp <- as.matrix(fit_p); dnb <- as.matrix(fit_nb)
rbind(Poisson = quantile(exp(dp[, "a1"]), c(0.5, 0.025, 0.975)),
  NegBin = quantile(exp(dnb[, "a1"]), c(0.5, 0.025, 0.975)))

```

	50%	2.5%	97.5%
Poisson	0.7109305	0.6335773	0.7962021
NegBin	0.7089796	0.5762211	0.8912823

The point estimates are similar, but the Poisson interval is much narrower: it treats the data as more informative than they are.

(c)

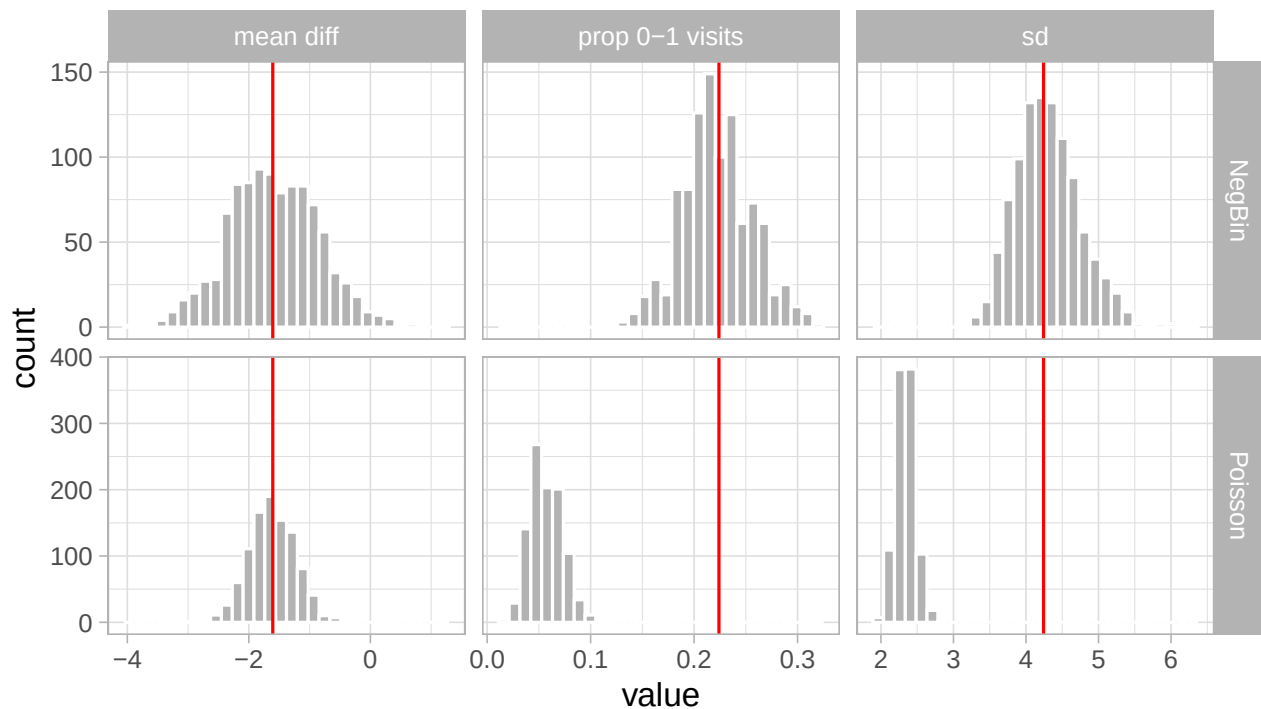
```

Tstats <- function(y) c(sd = sd(y), `prop 0-1 visits` = mean(y <= 1),
                        `mean diff` = mean(y[visits_df$far == 1]) -
                        ↪ mean(y[visits_df$far == 0]))
T_obs <- Tstats(visits_df$visits)

set.seed(53)
ppc <- function(dr, nb) {
  map_dfr(sample(nrow(dr), 1000), function(s) {
    mu <- exp(dr[s, "a0"] + dr[s, "a1"] * visits_df$far)
    yrep <- if (nb) rlnbinom(M, size = 1 / dr[s, "phi"], mu = mu) else rpois(M, mu)
    as_tibble_row(Tstats(yrep))
  })
}
ppc_all <- bind_rows(Poisson = ppc(dp, FALSE), NegBin = ppc(dnb, TRUE), .id =
  ↪ "model") |>
  pivot_longer(-model, names_to = "stat")

ggplot(ppc_all, aes(value)) + geom_histogram(bins = 30, fill = "grey70", colour =
  ↪ "white") +
  geom_vline(data = enframe(T_obs, "stat", "obs"), aes(xintercept = obs), colour
  ↪ = "red") +
  facet_grid(model ~ stat, scales = "free")

```



```

ppc_all |> left_join(enframe(T_obs, "stat", "obs"), by = "stat") |>
  group_by(model, stat) |> summarise(ppp = mean(value >= obs), .groups = "drop")
  ↪ |> kable(digits = 3)

```

model	stat	ppp
NegBin	mean diff	0.501
NegBin	prop 0-1 visits	0.486
NegBin	sd	0.525
Poisson	mean diff	0.477
Poisson	prop 0-1 visits	0.000
Poisson	sd	0.000

The Poisson model fails the SD and “0-1 visits” checks badly (posterior predictive p-values of 0). It does fine on the mean difference: a model can reproduce the feature you are interested in while getting the uncertainty wrong. The negative binomial passes all three checks.

(d)

```
ll_p <- matrix(dpois(matrix(visits_df$visits, nrow(dp), M, byrow = TRUE),
  exp(dp[, "a0"] + outer(dp[, "a1"], visits_df$far)), log =
  ↪ TRUE), nrow(dp))
ll_nb <- matrix(dnbinom(matrix(visits_df$visits, nrow(dnb), M, byrow = TRUE), size
  ↪ = 1 / dnb[, "phi"],
  mu = exp(dnb[, "a0"] + outer(dnb[, "a1"], visits_df$far)),
  ↪ log = TRUE), nrow(dnb))
loo_compare(list(Poisson = loo_from_ll(ll_p), NegBin = loo_from_ll(ll_nb)))
```

	elpd_diff	se_diff
NegBin	0.0	0.0
Poisson	-140.7	28.2

The negative binomial model has much higher expected log predictive density (a difference of many standard errors). Report the negative binomial model: it is supported by both the posterior predictive checks and LOO, and its wider interval for the rate ratio honestly reflects the variability in the data.

Exercise 3: a hierarchical model for multi-site data

HIV testing uptake among antenatal clinic attendees at 12 health centres (numbers offered a test and numbers accepting):

```
hc <- tibble(centre = sprintf("HC%02d", 1:12),
             offered = c(220, 35, 140, 12, 310, 60, 88, 25, 190, 8, 150, 45),
             accepted = c(198, 26, 131, 12, 270, 55, 71, 22, 176, 5, 139, 37))
hc |> mutate(raw = accepted / offered)
```

```
# A tibble: 12 x 4
  centre offered accepted  raw
  <chr>    <dbl>    <dbl> <dbl>
1 HC01      220      198 0.9
2 HC02       35       26 0.743
3 HC03      140      131 0.936
4 HC04       12       12 1
5 HC05      310      270 0.871
6 HC06       60       55 0.917
7 HC07       88       71 0.807
8 HC08       25       22 0.88
9 HC09      190      176 0.926
10 HC10       8        5 0.625
11 HC11      150      139 0.927
12 HC12       45       37 0.822
```

- Fit three models for the centre-specific acceptance probabilities p_j : complete pooling (one common p), no pooling (independent $p_j \sim \text{Beta}(1, 1)$; you may use conjugacy rather than MCMC), and partial pooling with $\text{logit}(p_j) = \mu + \sigma z_j$, $z_j \sim \mathcal{N}(0, 1)$, $\mu \sim \mathcal{N}(1.5, 1.5^2)$, $\sigma \sim \text{Exp}(1)$.
- Plot the three sets of estimates (with 95% intervals for no and partial pooling) against the centre sample size. Which centres are shrunk the most?
- The district target is 85% uptake. For each centre, compute $P(p_j < 0.85 \mid \text{data})$ under no pooling and under partial pooling. Which centres would you flag for supportive supervision under each model? Comment on HC04 and HC10.
- Refit the hierarchical model with the *centred* parameterisation, $\text{logit}(p_j) \sim \mathcal{N}(\mu, \sigma^2)$. Compare $\hat{R}$ and ESS for σ between the two parameterisations.

Solution

- and (b)

```

hier_nc <- nimbleCode({
  for (j in 1:J) {
    y[j] ~ dbin(p[j], n[j])
    logit(p[j]) <- mu + sigma * z[j]
    z[j] ~ dnorm(0, sd = 1)
  }
  mu ~ dnorm(1.5, sd = 1.5)
  sigma ~ dexp(1)
})

```

```

J <- nrow(hc)
fit_nc <- nimbleMCMC(hier_nc, constants = list(J = J, n = hc$offered), data =
  ↪ list(y = hc$accepted),
  inits = function() list(mu = rnorm(1, 1.5, 0.5), sigma = runif(1, 0.2, 1), z =
    ↪ rnorm(J)),
  monitors = c("mu", "sigma", "p"), nchains = 4, niter = 11000, nburnin = 1000,
  setSeed = TRUE, progressBar = FALSE, samplesAsCodaMCMC = TRUE)

```

```

dh <- as.matrix(fit_nc); pj <- dh[, paste0("p[", 1:J, "]")]
pooled <- sum(hc$accepted) / sum(hc$offered)

```

```

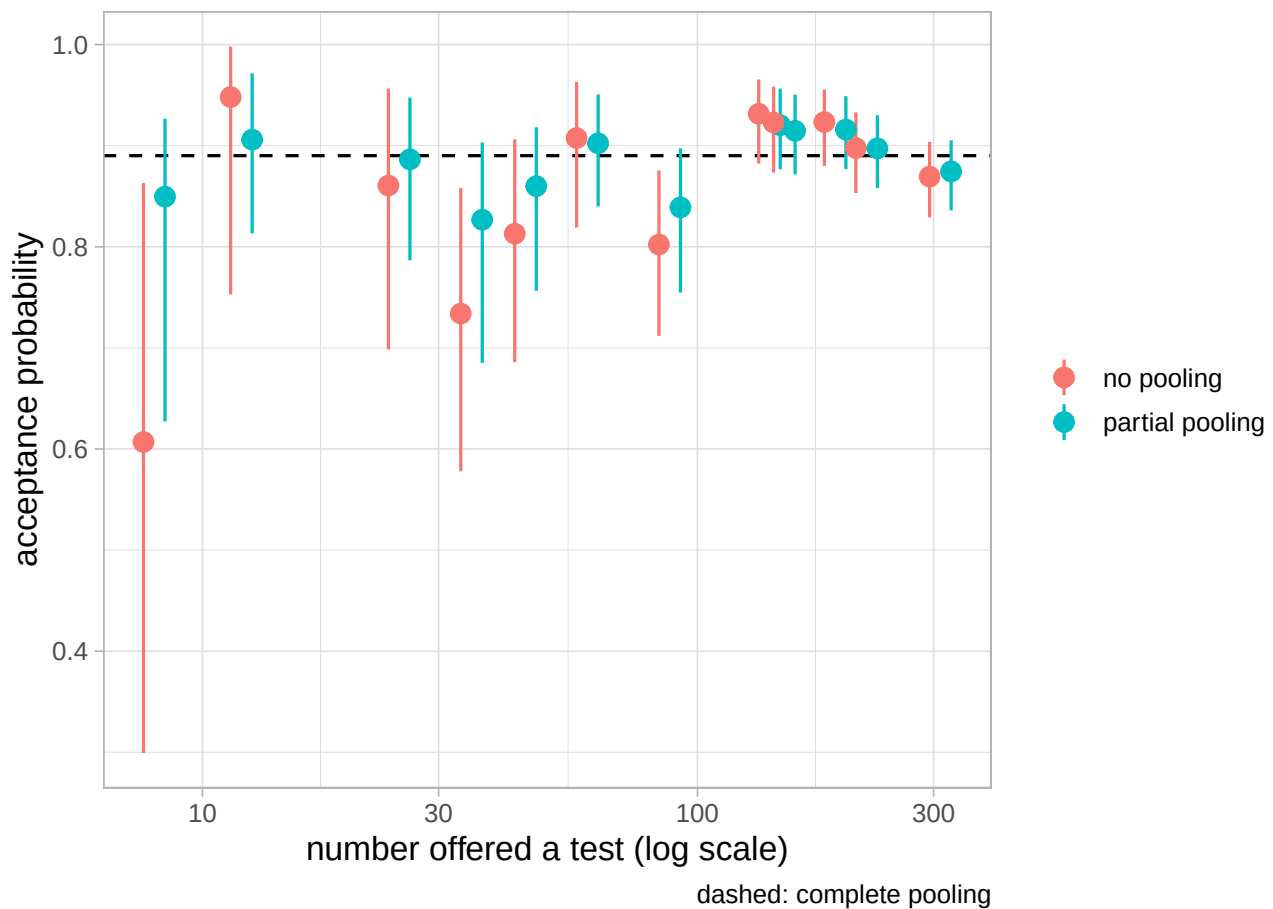
est <- hc |>
  mutate(np_med = qbeta(0.5, 1 + accepted, 1 + offered - accepted),
         np_lo = qbeta(0.025, 1 + accepted, 1 + offered - accepted),
         np_hi = qbeta(0.975, 1 + accepted, 1 + offered - accepted),
         pp_med = apply(pj, 2, median),
         pp_lo = apply(pj, 2, quantile, 0.025),
         pp_hi = apply(pj, 2, quantile, 0.975))

```

```

est |>
  select(centre, offered, starts_with("np"), starts_with("pp")) |>
  pivot_longer(-c(centre, offered), names_to = c("model", ".value"), names_sep =
    ↪ "_") |>
  mutate(model = recode(model, np = "no pooling", pp = "partial pooling"),
         x = offered * ifelse(model == "no pooling", 0.95, 1.05)) |> # small
    ↪ offset so intervals don't overlap
  ggplot(aes(x, med, colour = model)) +
  geom_hline(yintercept = pooled, linetype = 2) +
  geom_pointrange(aes(ymin = lo, ymax = hi)) +
  scale_x_log10() + labs(x = "number offered a test (log scale)", y = "acceptance
    ↪ probability",
                        caption = "dashed: complete pooling", colour = NULL)

```



Shrinkage towards the overall mean is strongest for the smallest centres (HC10 with 8 offered, HC04 with 12, HC08 with 25); the large centres hardly move.

(c)

```
est |>
  mutate(`P(p<0.85), no pooling` = pbeta(0.85, 1 + accepted, 1 + offered -
    ↪ accepted),
    `P(p<0.85), partial pooling` = colMeans(pj < 0.85)) |>
  select(centre, offered, accepted, `P(p<0.85), no pooling`, `P(p<0.85), partial
    ↪ pooling`) |>
  kable(digits = 3)
```

centre	offered	accepted	P(p<0.85), no pooling	P(p<0.85), partial pooling
HC01	220	198	0.018	0.010
HC02	35	26	0.965	0.668
HC03	140	131	0.001	0.001
HC04	12	12	0.121	0.085
HC05	310	270	0.164	0.097
HC06	60	55	0.089	0.045
HC07	88	71	0.888	0.617

centre	offered	accepted	P(p<0.85), no pooling	P(p<0.85), partial pooling
HC08	25	22	0.439	0.189
HC09	190	176	0.001	0.001
HC10	8	5	0.966	0.503
HC11	150	139	0.003	0.003
HC12	45	37	0.754	0.404

Under no pooling HC10 (5/8) looks almost certainly below target ($P \approx 0.97$), and HC04 (12/12) looks excellent. Partial pooling tempers both conclusions: with so few clients their extreme proportions are weak evidence, and the model borrows strength from the other centres, so HC10's probability of being below target drops to about 0.5. HC02 and HC07, which have more clients and low observed uptake, remain the centres most likely to be below target under partial pooling (about 0.6-0.7), though with less certainty than under no pooling. Supervision might sensibly start with HC02 and HC07; for HC10 the right decision is to collect more data rather than act on noise.

(d)

```
hier_c <- nimbleCode({
  for (j in 1:J) {
    y[j] ~ dbin(p[j], n[j])
    logit(p[j]) <- eta[j]
    eta[j] ~ dnorm(mu, sd = sigma)
  }
  mu ~ dnorm(1.5, sd = 1.5)
  sigma ~ dexp(1)
})
```

```
fit_c <- nimbleMCMC(hier_c, constants = list(J = J, n = hc$offered), data = list(y
  ↪ = hc$accepted),
  inits = function() list(mu = rnorm(1, 1.5, 0.5), sigma = runif(1, 0.2, 1), eta
  ↪ = rnorm(J, 1.5, 0.5)),
  monitors = c("mu", "sigma", "p"), nchains = 4, niter = 11000, nburnin = 1000,
  setSeed = TRUE, progressBar = FALSE, samplesAsCodaMCMC = TRUE)
```

```
bind_rows(`non-centred` = summarise_draws(as_draws(fit_nc), "rhat", "ess_bulk",
  ↪ "ess_tail"),
  centred = summarise_draws(as_draws(fit_c), "rhat", "ess_bulk",
  ↪ "ess_tail"),
  .id = "parameterisation") |>
  filter(variable %in% c("mu", "sigma")) |> kable(digits = 3)
```

parameterisation	variable	rhat	ess_bulk	ess_tail
non-centred	mu	1.002	1036.379	1598.131
non-centred	sigma	1.004	1527.035	2334.475

parameterisation	variable	rhat	ess_bulk	ess_tail
centred	mu	1.002	6639.686	10955.995
centred	sigma	1.008	331.806	125.849

Both parameterisations define the same model. Which one samples better depends on how informative the data are about each group: with small groups and a small σ , the centred version tends to have much lower ESS for σ (the “funnel”); with large, informative groups the centred version can be better. Always check the diagnostics, and try the other parameterisation if σ mixes poorly.